

HARSHITHA PUTTASWAMY

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PROFESSIONAL EXPERIENCE

Research Assistant, Scalable LLM Inference Framework for Fairness Evaluation

February 2025 – Present

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Built a scalable framework to analyze fairness in item allocations using LLMs, reducing manual evaluation and speeding up comparisons across algorithms.
- Boosted model inference speed 4x by running LLMs in parallel via distributed Ollama instances and multiple GPUs.
- Led prompt engineering strategies to elicit agent preferences, integrating statistical tests to analyze independence between algorithms and personality-driven behavior.

Research Assistant, Reinforcement Learning & Algorithmic Optimization

January 2024 – December 2024

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Designed and implemented a dynamic Multi-Armed Bandit (MAB) framework, creatively modeling voters as agents and candidates as arms, to optimize iterative voting decisions in multi-agent committee selection.
- Engineered scalable Integer Linear Programming (ILP) algorithms, leveraging the Gurobi solver, to efficiently solve complex multi-agent resource allocation and ranking problems.
- Conducted extensive experimental analysis, generating actionable insights for maximizing utility and satisfaction, applicable to designing real-time bidding environments.

Backend Developer, Internship

March 2022 – December 2022

Fyle Technologies Pvt Ltd

Bangalore, India

- Developed and optimized RESTful APIs using Django, Angular, and PostgreSQL (PSQL), and collaborated with project managers to implement 8 new features that increased market competitiveness and user engagement.
- Spearheaded comprehensive CI/CD pipeline implementation using Pytest and GitHub Actions, achieving over 98% code coverage, improving code quality for 3 core modules and reducing code review time by 90%
- Authored and delivered session on Pytest, GitHub Actions, and CI principles, empowering team adoption of best practices.

PROJECT EXPERIENCE

Immigration Law Chatbot (Fine-Tuned LLM with RAG on AWS) | AWS, Mistral-7B, RAG

May 2025 – June 2025

- Developed and fine-tuned a Mistral-7B-instruct LLM on AWS SageMaker for legal Q&A, leveraging a 9,000+ pair custom dataset.
- Architected and deployed a production-ready, scalable LLM inference system on AWS Lambda and API Gateway, significantly reducing hallucination and ensuring answer relevance by integrating Retrieval-Augmented Generation (RAG) via Amazon OpenSearch.
- Engineered robust data pipelines for LLM training and RAG knowledge base, ensuring reliable and seamless ingestion and retrieval of large datasets and source documents from AWS S3.

Splitwiser – A Splitwise Clone | Full Stack Development, Amazon Bedrock

June 2024 – December 2024

- Developed a full-stack application with receipt scanning, comprehensive group and individual expense tracking, and dynamic split functionalities for enhanced user experience.
- Leveraged Amazon Bedrock API to convert receipt images into JSON data, Next.js for responsive UI and API development, Google Firebase for robust backend services and Firestore for scalable receipt data storage.

Autokube – Kubernetes Cluster Automation Platform | DevOps, CI/CD

August 2021 – March 2022

- Led the end-to-end development of a web application designed to automate Kubernetes cluster creation and modification, streamlining DevOps workflows.
- Developed robust APIs using the MERN stack (MongoDB, Express.js, React, Node.js) to programmatically initiate scripts, orchestrating complex workflows for efficient management of backend Kubernetes clusters.

EDUCATION

Master of Science in Computer Science

January 2023 – December 2024

Binghamton University, SUNY (State University of New York)

Binghamton, NY, USA

Coursework: Deep Learning, High Performance Computing, Computer Architecture, Design and Analysis of Algorithms

TECHNICAL SKILLS

Programming Languages: C, CPP, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MySQL, Firebase

Frameworks and Libraries: TensorFlow, Keras, PyTorch, CUDA, OpenMP, MPI (familiarity via HPC coursework)

Cloud and Tools: AWS, Git, Linux, Kubernetes, GitHub Actions, Pytest

ML and AI: Recommendation Systems, Pattern Recognition, Deep Learning, Reinforcement Learning, NLP, LLM

Core Computer Science: Data Structures, Algorithm Design & Analysis, Distributed Systems, Computer Architecture

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate

- Validates expertise in building, deploying, and managing machine learning solutions on AWS